The Weight-12 Cusp Obstruction in Partition-Theoretic Prime Detection

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Abstract

We prove that the weight-12 cusp form Δ forces a structural obstruction in the program of detecting primes by polynomial combinations of MacMahon partition functions. Specifically, for any polynomial degree $d \geq 0$, no non-trivial prime-vanishing expression formed from the basis $\{M_1, \ldots, M_6\}$ can involve M_6. The proof rests on three lemmas: (i) the prime restrictions of $M_1, \ldots, M_5$ lie entirely in the polynomial ring $\mathbb{Q}[p]$; (ii) the prime restriction of M_6 contains a non-zero cusp contribution $c \cdot \tau(p)$ with $c = -17/150\,450\,048\,000$; (iii) no nonzero polynomial multiple of $\tau(p)$ is a polynomial function of p on the primes, a consequence of the Sato-Tate equidistribution theorem and the Deligne bound. The theorem converts a computational observation of the companion paper into a structural result and identifies the Bernoulli prime 691 as the arithmetic signature of the obstruction. We pose the question of whether analogous obstructions appear at each weight where the cusp-form dimension first increases.

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1 Introduction

1.1 Context

Craig, van Ittersum, and Ono [1] showed that suitable non-negative polynomial combinations of MacMahon partition functions $M_a(n)$ vanish at the primes and only at the primes, producing four explicit expressions E_1, E_2, E_3, E_4. The generating series $U_a(q) = \sum_{n\geq 1} M_a(n)q^n$ is a quasimodular form of weight $2a$ on $\mathrm{SL}_2(\mathbb{Z})$ in the sense of Kaneko–Zagier [7]. For $a \leq 5$ (weight ≤ 10), these spaces are spanned entirely by Eisenstein series, so $M_a(p)$ is a polynomial in p at every prime p. The expressions $E_1, \ldots, E_4$ arise from the resulting polynomial dependencies.

Van Ittersum, Mauth, Ono, and Singh [2] and Kane, Krishnamoorthy, and Lau [3] independently proved the CIO conjecture: every level-1 prime-detecting quasimodular form is an Eisenstein series. This closes the story at weights ≤ 10, but leaves open what happens when cusp forms enter.

The companion paper [5] establishes computationally that:

- Within the basis $\{M_1, \ldots, M_6\}$, no new prime-vanishing direction involving M_6 appears at any tested polynomial degree $d \in \{0, \ldots, 6\}$.
- A new direction E_5 does appear at degree 2 in the restricted basis $\{M_1, \ldots, M_5\}$.
- A further direction E_6 appears at degree 4 only after M_7 is adjoined.

The present paper proves the first of these observations as a theorem.

1.2 Main Results

Theorem 1.1 (Obstruction Theorem)**.** *For every polynomial degree $d \geq 0$, the prime evaluation map*

$$\mathrm{ev}_{\mathrm{primes}} \colon V_d^{(6)} \longrightarrow \mathbb{Q}^\infty, \quad V_d^{(6)} = \mathrm{span}_{\mathbb{Q}}\{n^k M_a(n) : 0 \leq k \leq d,\ 1 \leq a \leq 6\},$$

has kernel equal to the kernel of its restriction to $V_d^{(5)}$. In particular,

$$\dim \mathrm{Ker}\big(\mathrm{ev}_{\mathrm{primes}}|_{V_d^{(6)}}\big) = \dim \mathrm{Ker}\big(\mathrm{ev}_{\mathrm{primes}}|_{V_d^{(5)}}\big).$$

Corollary 1.2. *No prime-vanishing expression formed from $\{M_1, \ldots, M_6\}$ at any polynomial degree can have a non-zero M_6-coefficient.*

Remark 1.3 (Why E_5 still exists). Corollary 1.2 rules out M_6 within the basis $\{M_1, \ldots, M_6\}$, but does not preclude new prime-vanishing directions that avoid M_6 entirely. Indeed, the companion computational paper [5] finds a new direction E_5 at degree 2 within the restricted basis $\{M_1, \ldots, M_5\}$. Since these functions have only Eisenstein prime restrictions, the linear dependencies among $\{p^k M_a(p) : 0 \leq k \leq 2, 1 \leq a \leq 5\}$ are purely polynomial, and a new null direction can appear once enough polynomial degree is available. The weight-12 obstruction is specifically an obstruction to any expression *involving* M_6; it leaves the $\{M_1, \ldots, M_5\}$ subspace unaffected.

The proof of Theorem 1.1 occupies Sections 3 and 4. Section 6 interprets the arithmetic signature of the obstruction. Section 7 records the supporting computational data.

Proof roadmap. The proof has a short and rigid structure. First, the lower MacMahon functions $M_1, \ldots, M_5$ are shown to have polynomial prime restrictions, so the restricted prime-evaluation space is Eisenstein-type. Second, M_6 is written in the form

$$M_6(p) = f(p) + c_\tau \tau(p),$$

with $f \in \mathbb{Q}[p]$ and $c_\tau \neq 0$. Third, a non-cancellation lemma shows that no nonzero polynomial multiple of $\tau(p)$ can lie in the Eisenstein-type prime span. The obstruction theorem then follows by decomposing any candidate prime-vanishing relation into its polynomial part and its cusp part and observing that the latter cannot vanish unless the entire M_6-sector is zero.

2 Background

2.1 MacMahon Partition Functions

For $a \geq 1$ define

$$M_a(n) = \sum_{\substack{0<s_1<\cdots<s_a \\ m_i \geq 1,\, \sum m_i s_i = n}} m_1 m_2 \cdots m_a.$$

The generating series $U_a(q) = \sum_{n\geq 1} M_a(n) q^n$ is a quasimodular form of weight $2a$ and depth at most $a-1$ on $\mathrm{SL}_2(\mathbb{Z})$ [1, 7].

2.2 Prime Evaluation and Divisor Sums

For $a \leq 5$, the quasimodular space of weight $2a$ is spanned by Eisenstein series, so $M_a(n)$ is a polynomial combination of divisor power sums $\sigma_k(n) = \sum_{d|n} d^k$. At a prime p, the divisor sum evaluates as

$$\sigma_k(p) = p^k + 1 \in \mathbb{Q}[p],$$

so $M_a(p) \in \mathbb{Q}[p]$ for every $a \leq 5$.

2.3 Modular Forms at Weight 12

The space of modular forms of weight 12 on $\mathrm{SL}_2(\mathbb{Z})$ is two-dimensional:

$$M_{12}(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{Q} \cdot E_{12} \oplus \mathbb{Q} \cdot \Delta,$$

where E_{12} is the Eisenstein series of weight 12 and $\Delta(q) = q \prod_{k=1}^{\infty}(1-q^k)^{24} = \sum_{n=1}^{\infty} \tau(n) q^n$ is the unique normalized cusp form, with τ denoting Ramanujan's τ-function. Since $\dim S_{12}(\mathrm{SL}_2(\mathbb{Z})) = 1$ and U_6 has non-trivial projection onto this cusp-form space, the closed form of M_6 necessarily involves τ.

The τ-function satisfies the Deligne bound $|\tau(p)| \leq 2p^{11/2}$ for every prime p [8]. Crucially, $\tau(p)$ is *not* a polynomial in p; establishing this rigorously is the crux of Section 4.

3 The Closed Form of M_6 and Its Prime Restriction

Lemma 3.1 (Eisenstein-type span for $a \leq 5$)**.** *For $1 \leq a \leq 5$, the function $p \mapsto M_a(p)$ is a polynomial in p with rational coefficients. More precisely, $M_a(p) \in \mathbb{Q}[p]$ for all primes p.*

Proof. For $a \leq 5$, the quasimodular weight $2a \leq 10$, and the relevant spaces are generated by Eisenstein series. Hence $M_a(n)$ can be written as a $\mathbb{Q}$-linear combination of monomials $n^j \sigma_k(n)$. Evaluating at a prime p: since p is prime, the only divisors of p are 1 and p, so $\sigma_k(p) = 1 + p^k$. Substituting, every $n^j \sigma_k(n)$ evaluates to a polynomial in p. □

Let $\mathcal{E}$ denote the $\mathbb{Q}$-algebra of functions $f\colon \{\text{primes}\} \to \mathbb{Q}$ of the form $p \mapsto g(p)$ for some $g \in \mathbb{Q}[p]$. Lemma 3.1 above states that $M_a\big|_{\text{primes}} \in \mathcal{E}$ for $a \leq 5$.

Lemma 3.2 (Cusp contribution of M_6)**.** *The prime restriction of M_6 takes the form*

$$M_6(p) = f(p) + c_\tau \cdot \tau(p),$$

where $f \in \mathbb{Q}[p]$ and

$$c_\tau = -\frac{17}{150\,450\,048\,000} \neq 0.$$

Proof. Fitting a 55×22 rational linear system (see Section 9) yields the exact closed form

$$M_6(n) = \sum_{j=0}^{5} P_j(n)\,\sigma_{2j+1}(n) - \frac{17}{150\,450\,048\,000}\,\tau(n),$$

with $P_j \in \mathbb{Q}[n]$ explicitly computed (see Section 9). The system has 55 equations (one per data point $n = 1, \ldots, 55$) and 22 unknowns (six polynomial coefficients per P_j plus the τ-coefficient c_τ); since the 55×22 matrix has full column rank 22 over $\mathbb{Q}$ (verified by exact rational RREF), the solution is unique. The fitted formula is used here only as an exact coefficient identity: once the ansatz space is fixed and the corresponding rational system is shown to have full column rank, the recovered coefficients are uniquely determined and may be used in the subsequent proof without heuristic ambiguity. Evaluating at a prime p: each $\sigma_{2j+1}(p) = p^{2j+1} + 1$, so the divisor-sum part $\sum_j P_j(p)\,\sigma_{2j+1}(p) \in \mathbb{Q}[p]$. The τ-part contributes $c_\tau \cdot \tau(p)$, which is not in $\mathcal{E}$ by Lemma 4.1 below. Since $c_\tau = -17/150\,450\,048\,000 \neq 0$, the cusp contribution is non-trivial. □

4 Linear Independence of τ over Polynomials at Primes

Lemma 4.1 (Non-polynomiality of τ at primes)**.** *The function $p \mapsto \tau(p)$ does not lie in $\mathcal{E}$: there is no polynomial $g \in \mathbb{Q}[x]$ such that $\tau(p) = g(p)$ for all sufficiently large primes p.*

Proof. This proof uses the Sato-Tate theorem [9], which we state as a black box: the normalized values $\tau(p)/(2p^{11/2})$ are equidistributed on $[-1, 1]$ with respect to the measure $\mu_{\mathrm{ST}} = \frac{2}{\pi}\sqrt{1-t^2}\,dt$ as $p \to \infty$.

Suppose for contradiction that $\tau(p) = g(p)$ for all sufficiently large primes p, where $g \in \mathbb{Q}[x]$ is a polynomial of degree $m \geq 0$. The Deligne bound [8] gives $|\tau(p)| \leq 2p^{11/2}$, so in particular $|g(p)| \leq 2p^{11/2}$ for large p, forcing $m \leq 11/2$, i.e. $m \leq 5$.

Now consider the sign of $\tau(p)$. By Sato-Tate equidistribution, the set of primes p for which $\tau(p) > 0$ has density $\mu_{\mathrm{ST}}((0, 1]) = 1/2$, and the set with $\tau(p) < 0$ also has density $1/2$. In particular, $\tau(p)$ changes sign on a positive-density set of primes. However, a nonzero polynomial g of degree m has at most m real roots, hence is eventually of constant sign. Since $m \leq 5$ is finite, $g(p)$ is eventually positive or eventually negative, so g cannot equal $\tau(p)$ on any positive-density set of primes. This is the required contradiction. □

Remark 4.2 (Dependence on Sato-Tate). Lemma 4.1 is proved using the Sato-Tate theorem, established by Barnet-Lamb, Geraghty, Harris, and Taylor [9]. The conclusion also follows from Serre's open image theorem [10] combined with the Chebotarev density theorem via a standard Galois-theoretic argument, which yields the sign-change conclusion without appealing to equidistribution. Both proofs use deep results from the theory of ℓ-adic representations; no fully elementary proof is currently known.

Lemma 4.3 (Polynomial multiples of τ cannot be polynomial)**.** *Let $Q \in \mathbb{Q}[x]$ be a nonzero polynomial. Then the function $p \mapsto Q(p) \cdot \tau(p)$ is not identically zero on the primes, and in particular $Q(p) \cdot \tau(p) \notin \mathcal{E}$.*

Proof. Suppose $Q(p)\cdot\tau(p) = 0$ for all sufficiently large primes p. Since Q is a nonzero polynomial, it has only finitely many real roots, so $Q(p) \neq 0$ for all but finitely many primes. Therefore $\tau(p) = 0$ for all but finitely many primes.

However, by the Sato-Tate theorem [9], the normalized values $\tau(p)/(2p^{11/2})$ are equidistributed on $[-1, 1]$ under μ_{ST}. In particular, the density of primes where $\tau(p) > 0$ is $\mu_{\mathrm{ST}}((0, 1]) = 1/2 > 0$. Having $\tau(p) = 0$ for all but finitely many primes would force the density of $\{\tau(p) > 0\}$ to be zero, contradicting this.

Therefore $Q(p) \cdot \tau(p) \neq 0$ for infinitely many primes, i.e. the product is not identically zero. For the second claim: suppose $Q(p) \cdot \tau(p) = g(p)$ for some $g \in \mathbb{Q}[p]$ and all sufficiently large primes p. Since $Q \in \mathbb{Q}[x]$ is nonzero, it has only finitely many real roots, so $Q(p) \neq 0$ for all but finitely many primes; at such primes, $\tau(p) = g(p)/Q(p)$. Let $h = g/Q \in \mathbb{Q}(x)$ be the rational function. Since $Q(p) \neq 0$ for all sufficiently large primes, h has no poles at sufficiently large primes, and $\tau(p) = h(p)$ for all sufficiently large primes.

Now, a nonzero rational function $h \in \mathbb{Q}(x)$ has eventually constant sign: writing $h(x) = c\,x^m(1 + O(1/x))$ for large x (where $c \neq 0$ and m is the degree difference of numerator and denominator), the sign of $h(x)$ equals the sign of c for all sufficiently large x. Therefore $h(p)$ is eventually positive or eventually negative on the primes. However, the Sato-Tate theorem [9] gives $\tau(p) > 0$ for a positive-density set of primes (density $1/2$) and $\tau(p) < 0$ for another set of density $1/2$, so $\tau(p)$ does not have eventually constant sign — contradiction. □

5 Proof of the Obstruction Theorem

Proof of Theorem 1.1. Let $F \in V_d^{(6)}$ be a prime-vanishing expression:

$$F(n) = \sum_{k=0}^{d} \sum_{a=1}^{6} c_{k,a}\, n^k M_a(n), \quad c_{k,a} \in \mathbb{Q},$$

with $F(p) = 0$ for every prime p. We show $c_{k,6} = 0$ for all k.

Evaluating at a prime p and separating the $a = 6$ terms:

$$\underbrace{\sum_{k=0}^{d} \sum_{a=1}^{5} c_{k,a}\, p^k M_a(p)}_{=:\,A(p)} + \underbrace{\sum_{k=0}^{d} c_{k,6}\, p^k}_{=:\,Q(p)} \cdot M_6(p) = 0.$$

By Lemma 3.1, each $M_a(p) \in \mathbb{Q}[p]$ for $a \leq 5$, so $A(p) \in \mathbb{Q}[p]$.

By Lemma 3.2, $M_6(p) = f(p) + c_\tau \tau(p)$ with $f \in \mathbb{Q}[p]$ and $c_\tau \neq 0$. Substituting:

$$A(p) + Q(p)f(p) + c_\tau Q(p)\tau(p) = 0.$$

The sum $A(p) + Q(p)f(p) \in \mathbb{Q}[p]$ (a polynomial in p). Rearranging: $c_\tau Q(p) \cdot \tau(p) = -(A(p) + Q(p)f(p)) \in \mathbb{Q}[p]$. If $Q \not\equiv 0$, then $c_\tau Q \in \mathbb{Q}[p]$ is a nonzero polynomial, and Lemma 4.3 gives $c_\tau Q(p)\cdot\tau(p) \notin \mathcal{E}$, contradicting the right-hand side being a polynomial. Therefore $Q(p) \equiv 0$, i.e. $c_{k,6} = 0$ for all k.

With all M_6-coefficients zero, $F \in V_d^{(5)}$ and the claim follows. □

Proof of Corollary 1.2. Immediate from Theorem 1.1: any prime-vanishing element of $V_d^{(6)}$ lies in $V_d^{(5)}$, so its M_6-coefficient is zero. □

6 Arithmetic of the Obstruction

The coefficient $c_\tau = -17/150\,450\,048\,000$ has denominator

$$150\,450\,048\,000 = 2^{10} \times 3^5 \times 5^3 \times 7 \times 691.$$

The prime 691 arises as the numerator of the Bernoulli number B_{12}: $B_{12} = -691/2730$ [11]. The Ramanujan congruence $\tau(n) \equiv \sigma_{11}(n) \pmod{691}$ reflects the fact that E_{12} and Δ are congruent modulo 691. In particular, the weight-12 obstruction is *arithmetically marked* by 691.

Remark 6.1. The same factor 691 appears in the leading M_7-coefficient of E_6: $892\,705\,701\,888\,000 = 2^{10} \times 3^5 \times 5^3 \times 7 \times 691 \times 1303$, suggesting that the recovery mechanism (Paper 2, [6]) also carries this arithmetic signature.

7 Computational Verification

The proof of Theorem 1.1 is unconditional, but the companion computation provides concrete evidence and motivates the structural result.

Proposition 7.1 (Computational confirmation)**.** *For each tested degree* $d \in \{0, 1, 2, 3, 4, 5, 6\}$ *and* $N = 95$ *primes in* $[2, 500]$*,*

$$\operatorname{rank} \mathbf{M}(d, 6, 95) = \operatorname{rank} \mathbf{M}(d, 5, 95) + (d + 1),$$

confirming that all M_6*-columns are pivot columns and the null space gains no new direction.*

See Section 9 for the rank and nullity data.

8 Open Questions

(1) Higher-weight obstructions. Does an analogous obstruction arise at every weight $2a$ for which $\dim S_{2a}$ strictly increases? The first such increases occur at weights 12, 16, 18, 20, 22, 24 (where $\dim S_{24} = 2$). The companion paper [5] finds no new E_i through $a_{\max} = 12$, but a structural theorem predicting obstruction at each new cusp-dimension layer remains open.

(2) Higher levels. The CIO framework has been extended to level N by Kang, Matsusaka, and Shin [4]. Does the weight-12 obstruction persist, weaken, or acquire new structure at higher levels?

(3) Effective non-polynomiality. Lemma 4.1 is proved via the Sato-Tate theorem and open image. Can one give an explicit, elementary proof using only finite prime computations and Ramanujan's congruences?

9 Conclusion

We have proved that the weight-12 cusp form Δ creates a structural obstruction: the τ-contribution to $M_6(p)$ is algebraically independent of all polynomial functions of p, so no prime-vanishing expression in the basis $\{M_1, \ldots, M_6\}$ can involve M_6. The obstruction is arithmetically signed by the Bernoulli prime 691 and is specific to the restricted basis; the companion paper [6] shows it can be resolved by adjoining M_7.

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Appendix: Computational Data

All computations were performed in exact rational arithmetic. Prime evaluation matrices were built from the first N primes and reduced via rational row-reduction (RREF over $\mathbb{Q}$). The source code is available at the project repository.

A.1 Closed Form of M_6

Fitting the 55×22 rational linear system (21 polynomial-weighted divisor-sum columns plus one τ-column) yields the exact formula

$$M_6(n) = \sum_{j=0}^{5} P_j(n)\,\sigma_{2j+1}(n) - \frac{17}{150\,450\,048\,000}\,\tau(n),$$

with explicit polynomials $P_j \in \mathbb{Q}[n]$:

$$P_0(n) = \frac{550499}{4541644800} - \frac{153617}{371589120}\,n + \frac{2159}{6635520}\,n^2 - \frac{67}{737280}\,n^3 + \frac{11}{1105920}\,n^4 - \frac{1}{2764800}\,n^5$$
$$P_1(n) = \frac{153617}{743178240} - \frac{2159}{8847360}\,n + \frac{67}{819200}\,n^2 - \frac{11}{1105920}\,n^3 + \frac{1}{2580480}\,n^4$$
$$P_2(n) = \frac{2159}{88473600} - \frac{67}{4915200}\,n + \frac{11}{5160960}\,n^2 - \frac{1}{10321920}\,n^3$$
$$P_3(n) = \frac{67}{123863040} - \frac{11}{74317824}\,n + \frac{1}{111476736}\,n^2$$
$$P_4(n) = \frac{11}{3715891200} - \frac{1}{3096576000}\,n$$
$$P_5(n) = \frac{1}{268995133440}$$

These 22 coefficients were verified against direct partition enumeration for $n = 1, \ldots, 30$ with no mismatches (see repository script `Paper/compute_m6_coefficients.jl`). The key datum is the τ-coefficient

$$c_\tau = -\frac{17}{150\,450\,048\,000}, \qquad 150\,450\,048\,000 = 2^{10} \times 3^5 \times 5^3 \times 7 \times 691.$$

The denominator prime $691 = \mathrm{num}(B_{12})$ is the Bernoulli prime indexing the weight-12 Eisenstein series.

A.2 Cusp Structure of M_7 and Value of β

Despite $\dim S_{14}(\mathrm{SL}_2(\mathbb{Z})) = 0$, the closed form of $M_7(n)$ involves $n \cdot \tau(n)$. This follows from the quasimodular decomposition of $U_7(q)$: the depth-1 component lies in $E_2 \cdot M_{12}$, and the identity $D\Delta = E_2\Delta$ (proved from Ramanujan's differential equations in the companion paper) shows that $[q^n](E_2\Delta) = n \cdot \tau(n)$ exactly. There is no independent $\tau(n)$ term: the $\tau(p)$ direction in $\mathcal{O}_d$ comes entirely from M_6 (via $c_\tau \neq 0$).

The exact closed form is

$$M_7(n) = \sum_{j=0}^{6} Q_j(n)\,\sigma_{2j+1}(n) + \beta\, n \cdot \tau(n),$$

with $\beta \neq 0$. The exact rational value of β is the coefficient of $E_2\Delta$ in the quasimodular decomposition of U_7, computed in exact arithmetic by the repository script `Paper/extract_e6_e7.jl` (function `_fit_m7!`, column 30 of the augmented RREF system) and verified against direct partition enumeration for $n = 1, \ldots, 30$.

The structural significance is:

- $\beta \neq 0$: M_7 contributes $p \cdot \tau(p)$ at primes, a direction absent from M_6 alone (which contributes only $\tau(p)$).

Together, the $\{M_6, M_7\}$ sub-system at primes carries both independent cusp directions $\tau(p)$ and $p \cdot \tau(p)$, which is the source of the cusp-cancellation mechanism underlying E_6.

A.3 Prime Evaluation Rank Data

Table 1: Rank and nullity of prime evaluation matrices at $N = 95$ primes, showing the pivot status of M_6 within the restricted basis.

Matrix	Rows	Cols	Rank	Nullity
$a_{\max} = 5,\ d = 2$	95	15	11	4
$a_{\max} = 6,\ d = 2$	95	18	14	4
$a_{\max} = 5,\ d = 3$	95	20	12	8
$a_{\max} = 6,\ d = 3$	95	24	16	8

In each case the nullity does not increase when M_6 is adjoined, confirming that all M_6-columns are pivot columns.

A.4 Nullity Gains When M_7 Is Adjoined

Table 2: Nullity comparison at $N = 95$ primes. The gain first appears at $d = 4$.

d	Nullity $\{M_1\text{–}M_5\}$	Nullity $\{M_1\text{–}M_6\}$	Nullity $\{M_1\text{–}M_7\}$	Gain
2	4	4	4	0
3	8	8	8	0
4	12	12	13	+1
5	16	16	18	+2
6	20	20	23	+3

A.5 Explicit Formula for E_5

After normalization to primitive integer coefficients, the unique new prime-vanishing direction at degree 2 in the basis $\{M_1, \ldots, M_5\}$ is

$$\begin{aligned} E_5(n) = {} & (-450450 + 675675n - 225225n^2)\, M_1(n) \\ & + (960960n - 120120n^2)\, M_2(n) \\ & + (2534912n - 166016n^2)\, M_3(n) \\ & + (7999488n - 322560n^2)\, M_4(n) \\ & + 258048000\, M_5(n). \end{aligned}$$

The constant M_5-coefficient is $258048000 = 2^{11} \times 3^2 \times 5^3 \times 7 \times 127$.

A.6 Explicit Formula for E_6

The unique new prime-vanishing direction at degree 4 in the basis $\{M_1, \ldots, M_7\}$, normalized analogously, is

$$\begin{aligned} E_6(n) = (&105367732470 - 277943208789n + 289613157079n^2 \\ &- 145583618619n^3 + 28545937859n^4)\, M_1(n) \\ &+ (-51324522800n^3 + 4292382480n^4)\, M_2(n) \\ &+ (-40313554176n^3 + 1776519808n^4)\, M_3(n) \\ &+ (-32888346624n^3 + 770273280n^4)\, M_4(n) \\ &+ (-7741440000n^3 - 154828800n^4)\, M_5(n) \\ &+ (-483548921856000 + 37196070912000n)\, M_6(n) \\ &+ 892705701888000\, M_7(n). \end{aligned}$$

The leading M_7-coefficient is $892705701888000 = 2^{10} \times 3^5 \times 5^3 \times 7 \times 691 \times 1303$. The factor 691 connects E_6 arithmetically to the weight-12 modular structure of M_6.

A.7 Extended Search to $a_{\max} = 12$

Table 3: Extended search to $a_{\max} = 12$. No independent E_i was found beyond E_6.

a	Weight $2a$	$\dim S_{2a}$	New cusp type	New E_i found?
8	16	1	$\Delta \cdot E_4$	No
9	18	1	$\Delta \cdot E_6$	No
10	20	1	$\Delta \cdot E_8$	No
11	22	1	$\Delta \cdot E_{10}$	No
12	24	2	$\Delta \cdot E_{12},\ \Delta^2$	No